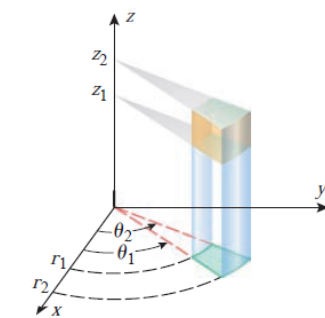


14.6 TRIPLE INTEGRALS IN CYLINDRICAL AND SPHERICAL COORDINATES

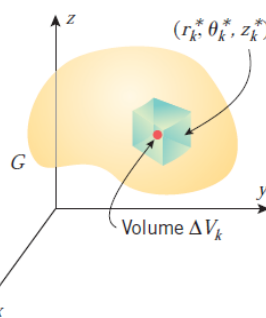
TRIPLE INTEGRALS IN CYLINDRICAL COORDINATES

Recall that in rectangular coordinates the triple integral of a continuous function f over a solid region G is defined as

$$\iiint_G f(x, y, z) dV = \lim_{n \rightarrow +\infty} \sum_{k=1}^n f(x_k^*, y_k^*, z_k^*) \Delta V_k$$



▲ Figure 14.6.1



▲ Figure 14.6.2

In cylindrical coordinates, the simplest equations are of the form

$$r = \text{constant}, \quad \theta = \text{constant}, \quad z = \text{constant}$$

The first equation represents a right circular cylinder centered on the z -axis, the second a vertical half-plane hinged on the z -axis, and the third a horizontal plane. (See Figure 11.8.3.) These surfaces can be paired up to determine solids called **cylindrical wedges** or **cylindrical elements of volume**. To be precise, a cylindrical wedge is a solid enclosed between six surfaces of the following form:

two cylinders (blue)	$r = r_1, \quad r = r_2 \quad (r_1 < r_2)$
two vertical half-planes (yellow)	$\theta = \theta_1, \quad \theta = \theta_2 \quad (\theta_1 < \theta_2)$
two horizontal planes (gray)	$z = z_1, \quad z = z_2 \quad (z_1 < z_2)$

(Figure 14.6.1). The dimensions $\theta_2 - \theta_1$, $r_2 - r_1$, and $z_2 - z_1$ are called the **central angle**, **thickness**, and **height** of the wedge.

$$dV = r \, dz \, dr \, d\theta$$

The region, E , over which we are integrating becomes,

$$\begin{aligned} E &= \{(x, y, z) \mid (x, y) \in D, \, u_1(x, y) \leq z \leq u_2(x, y)\} \\ &= \{(r, \theta, z) \mid \alpha \leq \theta \leq \beta, \, h_1(\theta) \leq r \leq h_2(\theta), \, u_1(r \cos \theta, r \sin \theta) \leq z \leq u_2(r \cos \theta, r \sin \theta)\} \end{aligned}$$

14.6.1 THEOREM Let G be a solid region whose upper surface has the equation $z = g_2(r, \theta)$ and whose lower surface has the equation $z = g_1(r, \theta)$ in cylindrical coordinates. If the projection of the solid on the xy -plane is a simple polar region R , and if $f(r, \theta, z)$ is continuous on G , then

$$\iiint_G f(r, \theta, z) dV = \iint_R \left[\int_{g_1(r, \theta)}^{g_2(r, \theta)} f(r, \theta, z) dz \right] dA \quad (4)$$

where the double integral over R is evaluated in polar coordinates. In particular, if the projection R is as shown in Figure 14.6.4, then (4) can be written as

$$\iiint_G f(r, \theta, z) dV = \int_{\theta_1}^{\theta_2} \int_{r_1(\theta)}^{r_2(\theta)} \int_{g_1(r, \theta)}^{g_2(r, \theta)} f(r, \theta, z) r dz dr d\theta \quad (5)$$

In terms of cylindrical coordinates a triple integral is,

$$\iiint_E f(x, y, z) dV = \int_{\alpha}^{\beta} \int_{h_1(\theta)}^{h_2(\theta)} \int_{u_1(r \cos \theta, r \sin \theta)}^{u_2(r \cos \theta, r \sin \theta)} r f(r \cos \theta, r \sin \theta, z) dz dr d\theta$$

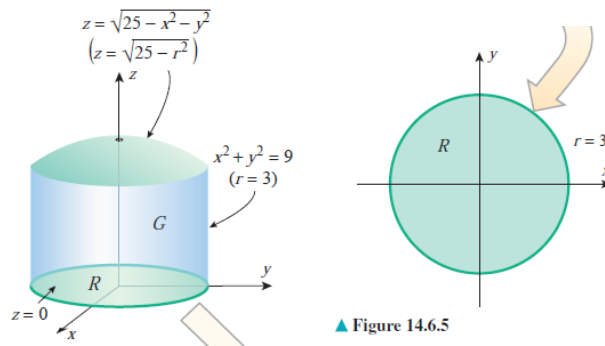
Determining Limits of Integration: Cylindrical Coordinates

- Step 1.** Identify the upper surface $z = g_2(r, \theta)$ and the lower surface $z = g_1(r, \theta)$ of the solid. The functions $g_1(r, \theta)$ and $g_2(r, \theta)$ determine the z -limits of integration. (If the upper and lower surfaces are given in rectangular coordinates, convert them to cylindrical coordinates.)
- Step 2.** Make a two-dimensional sketch of the projection R of the solid on the xy -plane. From this sketch the r - and θ -limits of integration may be obtained exactly as with double integrals in polar coordinates.

► **Example 1** Use triple integration in cylindrical coordinates to find the volume of the solid G that is bounded above by the hemisphere $z = \sqrt{25 - x^2 - y^2}$, below by the xy -plane, and laterally by the cylinder $x^2 + y^2 = 9$.

Solution. The solid G and its projection R on the xy -plane are shown in Figure 14.6.5. In cylindrical coordinates, the upper surface of G is the hemisphere $z = \sqrt{25 - r^2}$ and the lower surface is the plane $z = 0$. Thus, from (4), the volume of G is

$$V = \iiint_G dV = \iint_R \left[\int_0^{\sqrt{25-r^2}} dz \right] dA$$



▲ Figure 14.6.5

For the double integral over R , we use polar coordinates:

$$\begin{aligned} V &= \int_0^{2\pi} \int_0^3 \int_0^{\sqrt{25-r^2}} r \, dz \, dr \, d\theta = \int_0^{2\pi} \int_0^3 [rz]_{z=0}^{\sqrt{25-r^2}} dr \, d\theta \\ &= \int_0^{2\pi} \int_0^3 r\sqrt{25-r^2} \, dr \, d\theta = \int_0^{2\pi} \left[-\frac{1}{3}(25-r^2)^{3/2} \right]_{r=0}^3 d\theta \\ &= \int_0^{2\pi} \frac{61}{3} d\theta = \frac{122}{3}\pi \quad \leftarrow \begin{array}{l} u = 25 - r^2 \\ du = -2r \, dr \end{array} \end{aligned}$$

CONVERTING TRIPLE INTEGRALS FROM RECTANGULAR TO CYLINDRICAL COORDINATES

Sometimes a triple integral that is difficult to integrate in rectangular coordinates can be evaluated more easily by making the substitution $x = r \cos \theta$, $y = r \sin \theta$, $z = z$ to convert it to an integral in cylindrical coordinates. Under such a substitution, a rectangular triple integral can be expressed as an iterated integral in cylindrical coordinates as

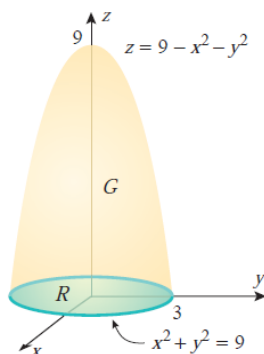
$$\iiint_G f(x, y, z) \, dV = \iiint_{\text{appropriate limits}} f(r \cos \theta, r \sin \theta, z) r \, dz \, dr \, d\theta \quad (6)$$

► **Example 2** Use cylindrical coordinates to evaluate

$$\int_{-3}^3 \int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} \int_0^{9-x^2-y^2} x^2 \, dz \, dy \, dx$$

Solution. In problems of this type, it is helpful to sketch the region of integration G and its projection R on the xy -plane. From the z -limits of integration, the upper surface of G is the paraboloid $z = 9 - x^2 - y^2$ and the lower surface is the xy -plane $z = 0$. From the x - and y -limits of integration, the projection R is the region in the xy -plane enclosed by the circle $x^2 + y^2 = 9$ (Figure 14.6.6). Thus,

$$\begin{aligned} \int_{-3}^3 \int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} \int_0^{9-x^2-y^2} x^2 \, dz \, dy \, dx &= \iiint_G x^2 \, dV \\ &= \iint_R \left[\int_0^{9-r^2} r^2 \cos^2 \theta \, dz \right] dA = \int_0^{2\pi} \int_0^3 \int_0^{9-r^2} (r^2 \cos^2 \theta) r \, dz \, dr \, d\theta \\ &= \int_0^{2\pi} \int_0^3 \int_0^{9-r^2} r^3 \cos^2 \theta \, dz \, dr \, d\theta = \int_0^{2\pi} \int_0^3 [zr^3 \cos^2 \theta]_{z=0}^{9-r^2} dr \, d\theta \\ &= \int_0^{2\pi} \int_0^3 (9r^3 - r^5) \cos^2 \theta \, dr \, d\theta = \int_0^{2\pi} \left[\left(\frac{9r^4}{4} - \frac{r^6}{6} \right) \cos^2 \theta \right]_{r=0}^3 d\theta \\ &= \frac{243}{4} \int_0^{2\pi} \cos^2 \theta \, d\theta = \frac{243}{4} \int_0^{2\pi} \frac{1}{2} (1 + \cos 2\theta) \, d\theta = \frac{243\pi}{4} \quad \blacktriangleleft \end{aligned}$$



▲ Figure 14.6.6

Example 1 Evaluate $\iiint_E y \, dV$ where E is the region that lies below the plane $z = x + 2$ above the xy -plane and between the cylinders $x^2 + y^2 = 1$ and $x^2 + y^2 = 4$.

Solution

There really isn't too much to do with this one other than do the conversions and then evaluate the integral.

We'll start out by getting the range for z in terms of cylindrical coordinates.

$$0 \leq z \leq x + 2 \quad \Rightarrow \quad 0 \leq z \leq r \cos \theta + 2$$

Remember that we are above the xy -plane and so we are above the plane $z = 0$

Next, the region D is the region between the two circles $x^2 + y^2 = 1$ and $x^2 + y^2 = 4$ in the xy -plane and so the ranges for it are,

$$0 \leq \theta \leq 2\pi \quad 1 \leq r \leq 2$$

Here is the integral.

$$\begin{aligned} \iiint_E y \, dV &= \int_0^{2\pi} \int_1^2 \int_0^{r \cos \theta + 2} (r \sin \theta) r \, dz \, dr \, d\theta \\ &= \int_0^{2\pi} \int_1^2 r^2 \sin \theta (r \cos \theta + 2) \, dr \, d\theta \\ &= \int_0^{2\pi} \int_1^2 \frac{1}{2} r^3 \sin(2\theta) + 2r^2 \sin \theta \, dr \, d\theta \\ &= \int_0^{2\pi} \left(\frac{1}{8} r^4 \sin(2\theta) + \frac{2}{3} r^3 \sin \theta \right) \Big|_1^2 \, d\theta \\ &= \int_0^{2\pi} \frac{15}{8} \sin(2\theta) + \frac{14}{3} \sin \theta \, d\theta \\ &= \left(-\frac{15}{16} \cos(2\theta) - \frac{14}{3} \cos \theta \right) \Big|_0^{2\pi} \\ &= 0 \end{aligned}$$

Example 2 Convert $\int_{-1}^1 \int_0^{\sqrt{1-y^2}} \int_{x^2+y^2}^{\sqrt{x^2+y^2}} xyz \, dz \, dx \, dy$ into an integral in cylindrical coordinates.

Solution

Here are the ranges of the variables from this iterated integral.

$$\begin{aligned} -1 &\leq y \leq 1 \\ 0 &\leq x \leq \sqrt{1-y^2} \\ x^2 + y^2 &\leq z \leq \sqrt{x^2 + y^2} \end{aligned}$$

The first two inequalities define the region D and since the upper and lower bounds for the x 's are $x = \sqrt{1-y^2}$ and $x = 0$ we know that we've got at least part of the right half a circle of radius 1 centered at the origin. Since the range of y 's is $-1 \leq y \leq 1$ we know that we have the complete right half of the disk of radius 1 centered at the origin. So, the ranges for D in cylindrical coordinates are,

$$\begin{aligned} -\frac{\pi}{2} &\leq \theta \leq \frac{\pi}{2} \\ 0 &\leq r \leq 1 \end{aligned}$$

$$r^2 \leq z \leq r$$

On a side note notice that the lower bound here is an elliptic paraboloid and the upper bound is a cone. Therefore, E is a portion of the region between these two surfaces.

The integral is,

$$\begin{aligned} \int_{-1}^1 \int_0^{\sqrt{1-y^2}} \int_{x^2+y^2}^{\sqrt{x^2+y^2}} xyz \, dz \, dx \, dy &= \int_{-\pi/2}^{\pi/2} \int_0^1 \int_{r^2}^r r(r \cos \theta)(r \sin \theta) z \, dz \, dr \, d\theta \\ &= \int_{-\pi/2}^{\pi/2} \int_0^1 \int_{r^2}^r zr^3 \cos \theta \sin \theta \, dz \, dr \, d\theta \end{aligned}$$

■ TRIPLE INTEGRALS IN SPHERICAL COORDINATES

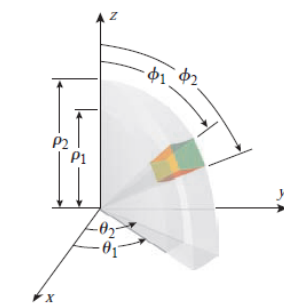
In spherical coordinates, the simplest equations are of the form

$$\rho = \text{constant}, \quad \theta = \text{constant}, \quad \phi = \text{constant}$$

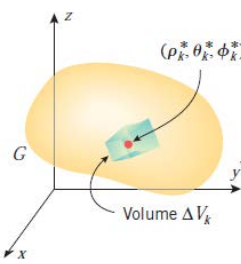
As indicated in Figure 11.8.4, the first equation represents a sphere centered at the origin and the second a half-plane hinged on the z -axis. The graph of the third equation is a right circular cone nappe with its vertex at the origin and its line of symmetry along the z -axis for $\phi \neq \pi/2$, and is the xy -plane if $\phi = \pi/2$. By a **spherical wedge** or **spherical element of volume** we mean a solid enclosed between six surfaces of the following form:

two spheres (green)	$\rho = \rho_1, \quad \rho = \rho_2 \quad (\rho_1 < \rho_2)$
two vertical half-planes (yellow)	$\theta = \theta_1, \quad \theta = \theta_2 \quad (\theta_1 < \theta_2)$
nappes of two circular cones (pink)	$\phi = \phi_1, \quad \phi = \phi_2 \quad (\phi_1 < \phi_2)$

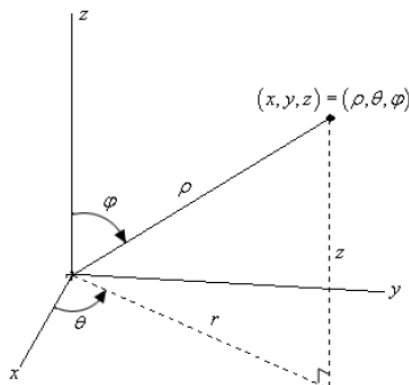
(Figure 14.6.7). We will refer to the numbers $\rho_2 - \rho_1$, $\theta_2 - \theta_1$, and $\phi_2 - \phi_1$ as the **dimensions** of a spherical wedge.



▲ Figure 14.6.7



▲ Figure 14.6.8



Here are the conversion formulas for spherical coordinates.

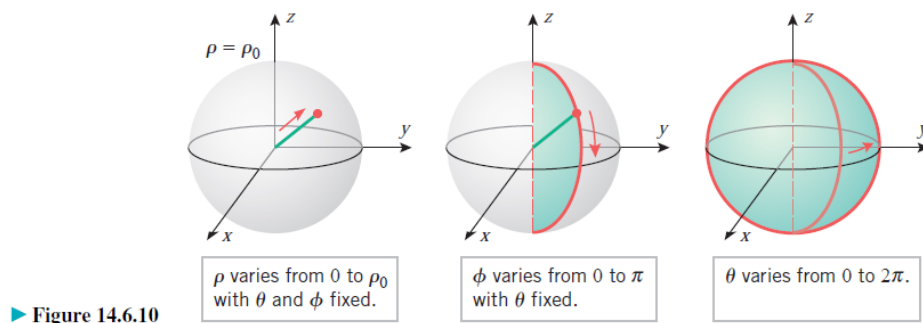
$$\begin{aligned} x &= \rho \sin \varphi \cos \theta & y &= \rho \sin \varphi \sin \theta & z &= \rho \cos \varphi \\ x^2 + y^2 + z^2 &= \rho^2 \end{aligned}$$

We also have the following restrictions on the coordinates.

$$\rho \geq 0 \quad 0 \leq \varphi \leq \pi$$

Suppose that we want to integrate $f(\rho, \theta, \phi)$ over the spherical solid G enclosed by the sphere $\rho = \rho_0$. The basic idea is to choose the limits of integration so that every point of the solid is accounted for in the integration process. Figure 14.6.10 illustrates one way of doing this. Holding θ and ϕ fixed for the first integration, we let ρ vary from 0 to ρ_0 . This covers a radial line from the origin to the surface of the sphere. Next, keeping θ fixed, we let ϕ vary from 0 to π so that the radial line sweeps out a fan-shaped region. Finally, we let θ vary from 0 to 2π so that the fan-shaped region makes a complete revolution, thereby sweeping out the entire sphere. Thus, the triple integral of $f(\rho, \theta, \phi)$ over the spherical solid G can be evaluated by writing

$$\iiint_G f(\rho, \theta, \phi) dV = \int_0^{2\pi} \int_0^\pi \int_0^{\rho_0} f(\rho, \theta, \phi) \rho^2 \sin \phi d\rho d\phi d\theta$$



$$dV = \rho^2 \sin \phi d\rho d\theta d\phi$$

Therefore, the integral will become,

$$\iiint_E f(x, y, z) dV = \int_{\delta}^{\gamma} \int_{\alpha}^{\beta} \int_a^b \rho^2 \sin \phi f(\rho \sin \phi \cos \theta, \rho \sin \phi \sin \theta, \rho \cos \phi) d\rho d\theta d\phi$$

Table 14.6.1 suggests how the limits of integration in spherical coordinates can be obtained for some other common solids.

Table 14.6.1

DETERMINATION OF LIMITS				INTEGRAL
				$\int_0^{\pi/2} \int_0^{\pi/2} \int_0^{\rho_0} f(\rho, \theta, \phi) \rho^2 \sin \phi d\rho d\phi d\theta$
ρ varies from 0 to ρ_0 with θ and ϕ held fixed.	ϕ varies from 0 to $\pi/2$ with θ held fixed.	θ varies from 0 to $\pi/2$.		

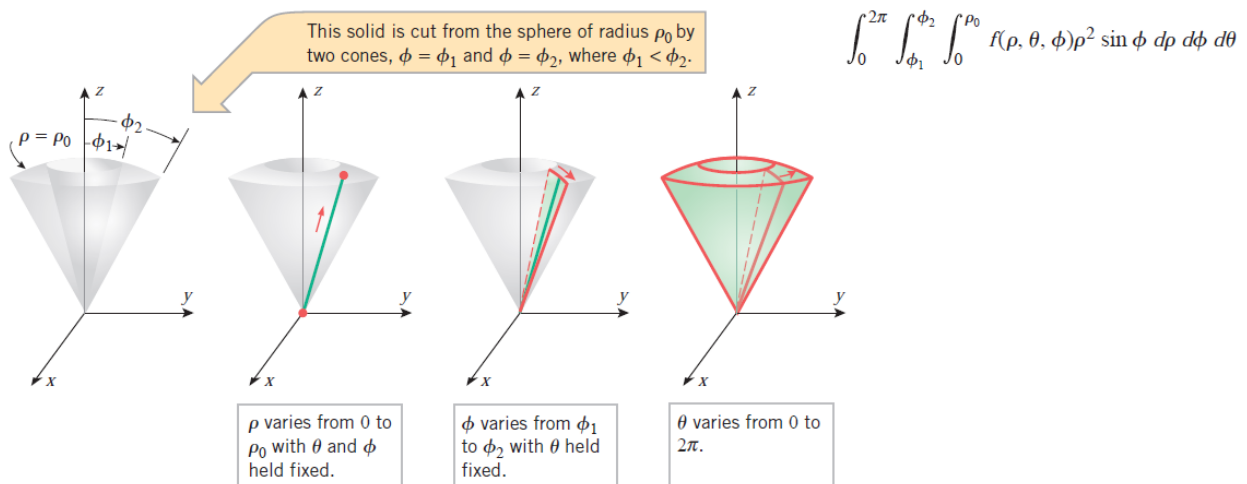
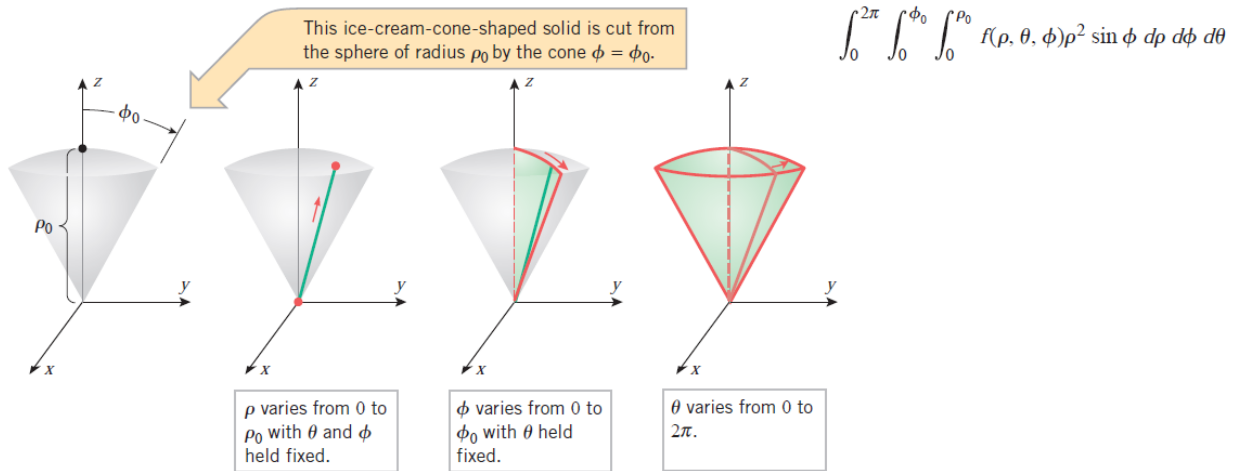
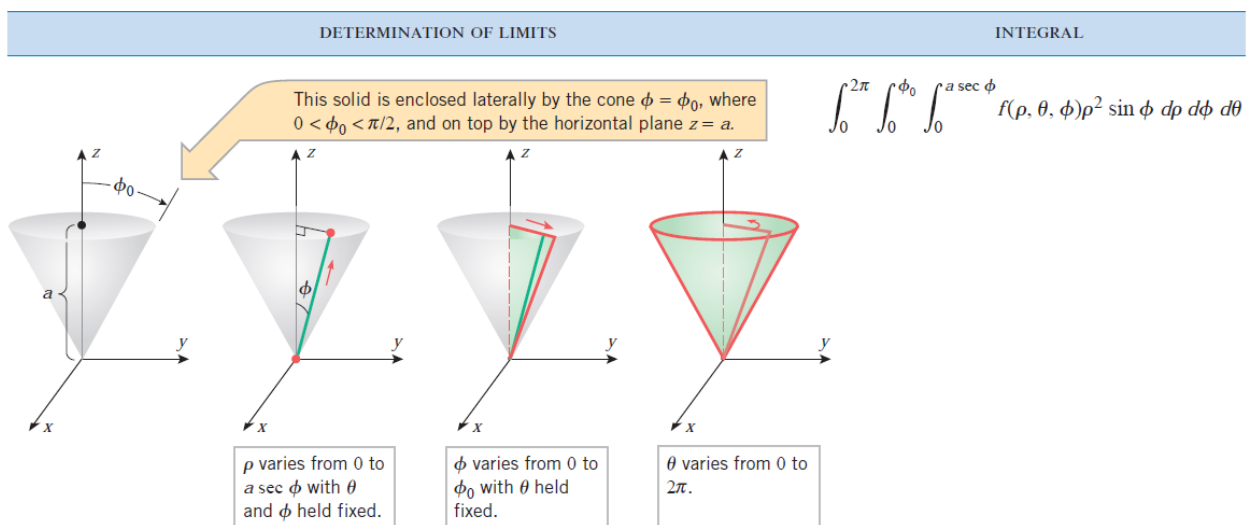
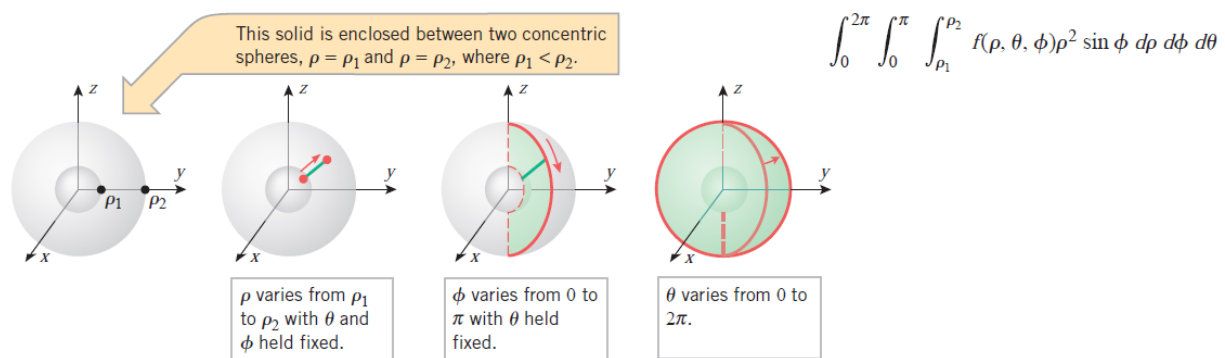


Table 14.6.1 (continued)





► **Example 3** Use spherical coordinates to find the volume of the solid G bounded above by the sphere $x^2 + y^2 + z^2 = 16$ and below by the cone $z = \sqrt{x^2 + y^2}$.

Solution. The solid G is sketched in Figure 14.6.11. In spherical coordinates, the equation of the sphere $x^2 + y^2 + z^2 = 16$ is $\rho = 4$ and the equation of the cone $z = \sqrt{x^2 + y^2}$ is

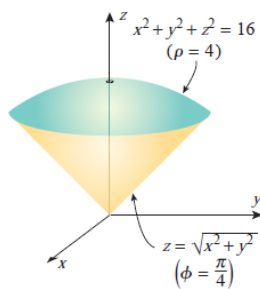
$$\rho \cos \phi = \sqrt{\rho^2 \sin^2 \phi \cos^2 \theta + \rho^2 \sin^2 \phi \sin^2 \theta}$$

which simplifies to

$$\rho \cos \phi = \rho \sin \phi$$

Dividing both sides of this equation by $\rho \cos \phi$ yields $\tan \phi = 1$, from which it follows that

$$\phi = \pi/4$$



▲ Figure 14.6.11

Thus, it follows from the second entry in Table 14.6.1 that the volume of G is

$$\begin{aligned}
 V &= \iiint_G dV = \int_0^{2\pi} \int_0^{\pi/4} \int_0^4 \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta \\
 &= \int_0^{2\pi} \int_0^{\pi/4} \left[\frac{\rho^3}{3} \sin \phi \right]_{\rho=0}^4 d\phi \, d\theta \\
 &= \int_0^{2\pi} \int_0^{\pi/4} \frac{64}{3} \sin \phi \, d\phi \, d\theta \\
 &= \frac{64}{3} \int_0^{2\pi} [-\cos \phi]_{\phi=0}^{\pi/4} d\theta = \frac{64}{3} \int_0^{2\pi} \left(1 - \frac{\sqrt{2}}{2} \right) d\theta \\
 &= \frac{64\pi}{3} (2 - \sqrt{2}) \approx 39.26 \quad \blacktriangleleft
 \end{aligned}$$

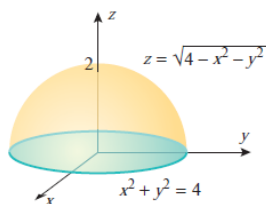
■ CONVERTING TRIPLE INTEGRALS FROM RECTANGULAR TO SPHERICAL COORDINATES

Referring to Table 11.8.1, triple integrals can be converted from rectangular coordinates to spherical coordinates by making the substitution $x = \rho \sin \phi \cos \theta$, $y = \rho \sin \phi \sin \theta$, $z = \rho \cos \phi$. The two integrals are related by the equation

$$\iiint_G f(x, y, z) \, dV = \iiint_{\text{appropriate limits}} f(\rho \sin \phi \cos \theta, \rho \sin \phi \sin \theta, \rho \cos \phi) \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta \quad (10)$$

► **Example 4** Use spherical coordinates to evaluate

$$\int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_0^{\sqrt{4-x^2-y^2}} z^2 \sqrt{x^2 + y^2 + z^2} \, dz \, dy \, dx$$



▲ Figure 14.6.12

Solution. In problems like this, it is helpful to begin (when possible) with a sketch of the region G of integration. From the z -limits of integration, the upper surface of G is the hemisphere $z = \sqrt{4 - x^2 - y^2}$ and the lower surface is the xy -plane $z = 0$. From the x - and y -limits of integration, the projection of the solid G on the xy -plane is the region enclosed by the circle $x^2 + y^2 = 4$. From this information we obtain the sketch of G in Figure 14.6.12. Thus,

$$\begin{aligned}
 & \int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_0^{\sqrt{4-x^2-y^2}} z^2 \sqrt{x^2 + y^2 + z^2} dz dy dx \\
 &= \iiint_G z^2 \sqrt{x^2 + y^2 + z^2} dV \\
 &= \int_0^{2\pi} \int_0^{\pi/2} \int_0^2 \rho^5 \cos^2 \phi \sin \phi d\rho d\phi d\theta \\
 &= \int_0^{2\pi} \int_0^{\pi/2} \frac{32}{3} \cos^2 \phi \sin \phi d\phi d\theta \\
 &= \frac{32}{3} \int_0^{2\pi} \left[-\frac{1}{3} \cos^3 \phi \right]_{\phi=0}^{\pi/2} d\theta = \frac{32}{9} \int_0^{2\pi} d\theta = \frac{64}{9} \pi \quad \blacktriangleleft
 \end{aligned}$$

Example 1 Evaluate $\iiint_E 16z dV$ where E is the upper half of the sphere $x^2 + y^2 + z^2 = 1$.

Solution

Since we are taking the upper half of the sphere the limits for the variables are,

$$0 \leq \rho \leq 1$$

$$0 \leq \theta \leq 2\pi$$

$$0 \leq \phi \leq \frac{\pi}{2}$$

$$\begin{aligned}
 \iiint_E 16z dV &= \int_0^{\pi/2} \int_0^{2\pi} \int_0^1 \rho^2 \sin \phi (16\rho \cos \phi) d\rho d\theta d\phi \\
 &= \int_0^{\pi/2} \int_0^{2\pi} \int_0^1 8\rho^3 \sin(2\phi) d\rho d\theta d\phi \\
 &= \int_0^{\pi/2} \int_0^{2\pi} 2 \sin(2\phi) d\theta d\phi \\
 &= \int_0^{\pi/2} 4\pi \sin(2\phi) d\phi \\
 &= -2\pi \cos(2\phi) \Big|_0^{\pi/2} \\
 &= 4\pi
 \end{aligned}$$

Example 2 Evaluate $\iiint_E z x dV$ where E is above $x^2 + y^2 + z^2 = 4$, inside the cone (pointing upward) that makes an angle of $\frac{\pi}{3}$ with the negative z-axis and has $x \leq 0$.

Solution

First, we need to take care of the limits. The region E is basically an upside down ice cream cone that has been cut in half so that only the portion with $x \leq 0$ remains. Therefore, because we are inside a portion of a sphere of radius 2 we must have,

$$0 \leq \rho \leq 2$$

For φ we need to be careful. The problem statement says that the cone makes an angle of $\frac{\pi}{3}$ with the negative z-axis. However, remember that φ is measured from the positive z-axis. Therefore, the first angle, as measured from the positive z-axis, that will “start” the cone will be $\varphi = \frac{2\pi}{3}$ and it goes to the negative z-axis. Therefore, we get the following limits for φ .

$$\frac{2\pi}{3} \leq \varphi \leq \pi$$

Finally, for the θ we can use the fact that we are also told that $x \leq 0$. This means we are to the left of the y-axis and so the range of θ must be,

$$\frac{\pi}{2} \leq \theta \leq \frac{3\pi}{2}$$

$$\begin{aligned} \iiint_E z x dV &= \int_{\frac{2\pi}{3}}^{\pi} \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \int_0^2 (\rho \cos \varphi)(\rho \sin \varphi \cos \theta) \rho^2 \sin \varphi d\rho d\theta d\varphi \\ &= \int_{\frac{2\pi}{3}}^{\pi} \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \int_0^2 \rho^4 \cos \varphi \sin^2 \varphi \cos \theta d\rho d\theta d\varphi \\ &= \int_{\frac{2\pi}{3}}^{\pi} \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \frac{32}{5} \cos \varphi \sin^2 \varphi \cos \theta d\theta d\varphi \\ &= \int_{\frac{2\pi}{3}}^{\pi} -\frac{64}{5} \cos \varphi \sin^2 \varphi d\varphi \\ &= -\frac{64}{15} \sin^3 \varphi \Big|_{\frac{2\pi}{3}}^{\pi} \\ &= \frac{8\sqrt{3}}{5} \end{aligned}$$

Practice Problem:

Chapter-14.6:

Problems: 9-12,13-16,17-20

Thank you for your attention!